

BME 5303 Special Topics “Orthopedics”

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Office hours: Tuesday 12:30 PM
or by appointment
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Class hours: 2:00-3:15 PM Tuesday and Thursday

Place: Taubman Complex J332 or as directed

Course Content

This elective will cover the impact of engineering on orthopedics, surgery and other clinical practices. General topics will include: biomaterials, biochemistry, biomechanics, bioimaging, bioinstrumentation, physiology, anatomy, experimental methods and biostatistics.

Course instructional format

Students will complete practical experiences to investigate the technical challenges that face clinicians. This may include lectures, seminars, tours, class activities, clinical practicums and research or design projects to solve clinical problems with new technology or by applying existing technology in innovative ways. Students will be assessed based on daily summaries, project assignments, research reports and presentations. This is a three credit class consisting of two 75 min lectures per week.

Recommended Textbook

Orthopaedic Basic Science 5th Edition, T.A. Einhorn, R.J. O’Keefe, J.A. Buckwalter.

American Academy of Orthopaedic Surgeons, 2020. ISBN: 0892033606

(There are newer versions of this book, but they are substantially more expensive. This is \$10 for paperback)

Grading

- **15% Attendance and Class Participation**
 - It is important that students attend class. Missing days will hurt your grade in this course. Please contact the instructor for any and all anticipated absences, and especially emergencies relating to class attendance. The student will get an “F” after the tenth unexcused absence.
- **15% Homework Assignments**
 - Students may discuss assignments, but should write up solutions independently. Assignments should be saved with all relevant source materials organized in a binder or digital folder. Late assignments will be penalized or not accepted, unless previously arranged with the instructor.
- **20% In Class Activities**
 - Students should work together as directed, but should write reflections independently. Assignments should be organized with all relevant materials in a course binder.
- **15% Journal Club**
 - Weekly reading and summaries to be included in binder. Assigned individual presentations on a scientific journal article to summarize and lead a discussion. Additional details will be provided during the semester.
- **20% Project**
 - Individual research and device design for an orthopedic topic of the students’ choosing. Additional details will be provided during the semester.
- **15% Final Exam**
 - One cumulative exam will assess the material covered during the course. The date and time of the exam is TBA.

Grading Scale:

Grades will be assigned as follows:

– A	≥ 93%
– A-	90-92.9
– B+	87-89.9%
– B	83-86.9%
– B-	80-82.9%
– C+	77-79.9%
– C	73-76.9%
– C-	70-72.9%
– D+	67-69.9%
– D	60-66.9%
– F	<60%

Attendance Rubric

Participation Grade for Semester	Perfect Attendance	2 Absence	4 Absences	6 Absences	8 Absences	10+ Absences
A	15%	15%	13%	11%	9%	Fail
B	13%	13%	11%	9%	7%	Fail
C	11%	11%	9%	7%	5%	Fail
D	9%	9%	7%	5%	3%	Fail
F	7%	7%	5%	3%	1%	Fail

A = Fully engaged in class and team discussions. Asked and answered questions from instructor and classmates.

B = Mostly engaged in class and team discussions. Occasionally asked and answered questions from instructor and classmates. Occasionally distracted or engaged in other activities.

C = Only partially engaged in class discussions. Only responded when directly asked a question by the instructor. Occasionally distracted or engaged in other activities.

D = Rarely participated in class discussions. Only partially able to answer questions directly posed by professor. Frequently distracted or engaged in other activities.

F = Attended class but did not actively participate. Unable to answer questions directly posed by professor.

For program assessment purposes, anonymous copies of submitted student work may be kept on file.

Violations of academic ethics will be handled according to University policies as set forth in the University's Academic Honor Code, included here by reference (see http://www.ltu.edu/currentstudents/honor_code.asp). Specific policies for assignments in this course are:

Note: Unforeseen circumstances may necessitate changes, additions, or deletions to this syllabus during the term.

Specific Topics: Depending on time and availability, not all topics may be covered or additional topics may be added

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| • Introduction to Orthopedics | • Regeneration |
| • ACL Injuries/ Reconstruction | • Total Joint Reconstruction |
| • Ligament properties | • Implant Failure |
| • Ankle Sprains | • Prosthetics/ Orthotics |
| • Cartilage properties/ Arthritis | • Spine |
| • Bone Fracture/ Implant Fixation | • Physical Therapy/ Injury Prevention |
| • Biomechanical Modeling | • Medical Imaging |
| • Custom Implants | • Head Injury |
| • Skeletal Development/ Osteochondral | • FDA Regulation of Orthopedic Devices |

Lawrence Technological University
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Preliminary schedule of topics and assignments for the Spring 2026 semester.

Date	Topic	In class (PSN due)	Presenter	Assignment
1/13/26	1 Introduction	Virtual Hip Surgery		HW1 THR Systems
1/15/26		Build Anatomy		HW2 Lower Extremity
1/20/26	2 Sports Injuries	Video Analysis	JC0	HW3 Xray/CT Analysis
1/22/26				Finish Lower Extremity
1/27/26	3 ACL Reconstruction		JC1	HW4 ACL Reconstruction
1/29/26		ACL Reconstruction		
2/3/26	4 Cartilage/Arthritis		JC2	HW5 Biomaterials
2/5/26		Hip Replacement		
2/10/26	5 Fracture Fixation		JC3	HW6 Fixation Devices
2/12/26		Fracture Fixation		
2/17/26	6 Spinal Fusion		JC4	<i>Project Topic Selection</i>
2/19/26		Spine Fixation		
2/24/26	7 Skeletal Development		JC5	HW7 Bone Biology
2/26/26		Materials Testing		Notebook Check
3/3/26	8 Implant Stability		JC6	
3/5/26		Dental Implants		<i>Project Abstract due</i>
3/17/26	9 Opportunity Creation		JC7	
3/19/26		Providence Tour		<i>Project Brainstorming</i>
3/24/26	10 FDA Regulation		JC8	<i>Project IP Search</i>
3/26/26		Commercialization		Project Regulatory/Quality Issues
3/31/26	11 Biomech Modeling		JC9	
4/2/26		Mimics		HW8 Computer Aided Design
4/7/26	12 Prosthetics/ Orthotics		JC10	HW9 Prosthetic Control
4/9/26		Tour?		
4/14/26	13 Injury Prevention		JC11	Project Report Due
4/16/26		Seminar?		
4/21/26	14 Medical Imaging		JC12	HW10 Overuse Injuries
4/23/26		Tour?		Take Home Final
4/28/26	15 Concussion		JC13	Notebooks Due
4/30/26		Seminar?		
TBD	Final Exam			In Class Presentations

Course Binder Assignment details:

- Due: April 28, 2026
- All in class activity summaries, JC summaries and homework assignments should be saved with all relevant source materials (save as .pdf files) and organized in a course binder or digital folder. Late assignments will be penalized or not accepted, unless previously arranged with the instructor. Details for each assignment are provided on Canvas and may also be submitted electronically for grading.